4635120

B.Tech. DEGREE EXAMINATION,
NOVEMBER/DECEMBER 2014.

Fifth Semester

Computer Science and Engineering

COMPUTER NETWORKS

Time : Three hours Maximum : 75 marks

PART A — ($10 \times 2 = 20$ marks)

Answer ALL questions.

All questions carry equal marks.

1. Differentiate connection oriented and connectionless service.

2. Mention the different types of physical media?

3. What is the function of SMTP?

4. Discuss the TCP connections needed in FTP.

5. What are the different types of multiplexing?

6. Give the format of HTTP response message.

7. Define ICMP.

8. What is meant by Router?

9. Give some application of Network.

10. What is the use of Hub?

PART B — (5 × 11 = 55 marks)

Answer ALL question, ONE from each Unit.

All questions carry equal marks.

UNIT I

11. Explain about the delay and loss in packet switched networks? (11)

Or

12. What is OSI Reference model? Explain the functions, protocols and services of each layer? (11)

UNIT II

13. Discuss about the Socket programming with UDP. (11)

Or

14. What is RTP and explain RTP packet header fields? (11)

UNIT III

15. Explain Multiplexing and Demultiplexing applications. (11)

Or

16. Write in detail about TCP Congestion Control. (11)

UNIT IV

17. Explain about the IPv4 datagram format? (11)

Or

18. (a) What is Multicast Routing? (3)

(b) Explain about the approaches used in that Routing. (8)

UNIT V

19. Write in detail about error detection and correction techniques. (11)

Or

20. Discuss about the Ethernet technologies. (11)

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B.Tech. DEGREE EXAMINATION, NOVEMBER 2017.

Fifth Semester

Computer Science and Engineering

COMPUTER NETWORKS

(2013-14 onwards)

Time : Three hours Maximum : 75 marks

PART A — (10 × 2 = 20 marks)

Answer ALL questions.

1. What are hotspots? Which protocol is used in hotspots to provide internet connectivity?

2. What is frequency hopping?

3. What is Carrier Sensing?

4. Differentiate switches and routers.

5. Differentiate broadcast and multicast routing.

6. Find the class of the following IP addresses

   (a) 172.16.16.201

   (b) 10.10.100.100

7. What are the advantages of TCP over UDP?

5. Define the term reverse path forwarding.

6. Draw IPv4 Header.

7. What is the use of portmapper?

8. Discuss the advantages and disadvantages of credits versus sliding window protocols.

9. Differentiate between recursive and iterative query.

10. What is reflection attack?

PART B — (5 × 11 = 55 marks)

Answer ALL Questions, ONE from each Unit.

All questions carry equal marks.

UNIT I

11. Explain the TCP/IP Architecture in detail.

Or

12. (a) Describe GEO MEO and LEO satellites. (5)

(b) Explain the transmission medium used in wireless communication. (6)

UNIT II

13. Explain sliding window protocol with suitable example.

Or

14. Describe CSMA and CSMA/CD protocols in detail.

UNIT III

15. Explain Distance vector routing protocol to solve count-to-infinity problem.

Or

16. Explain interior gateway routing protocol.

UNIT IV

17. Explain connection establishment and connection release in transport layer.

Or

18. Design and implement a chat system that allows multiple groups of users to chat. A chat coordinator resides at a well-known network address, uses UDP for communication with chat clients, sets up chat servers for each chat session, and maintains a chat session directory. There is one chat server per chat session. A chat server uses TCP for communication with clients. A chat client allows users to start, join, and leave a chat session. Design and implement the coordinator, server, and client code.

19. Explain induc

20. (a) Write t

(b) Explair

8. Draw the flow diagram for TCP connection termination.

9. What is URL? What is the significance of URL in HTTP?

10. What is a substitution cipher? Encrypt CRYPTOGRAPHY' using a Caesar cipher with key = 3.

PART B — (5 × 11 = 55 marks)

Answer ALL questions, ONE from each Unit.

11. (a) Explain in detail about spread spectrum communications. (6)

(b) Explain the physical properties of twisted pair and fiber optics wired transmission media. (5)

Or

12. (a) Explain in detail about Geostationary Satellite system. (6)

(b) What is meant by layered approach? List down the functionalities of data link and network layers in OSI reference model. (5)

13. (a) Explain in detail about CSMA protocols. (6)

(b) Explain about the protocol stack used in 802.16. (5)

Or

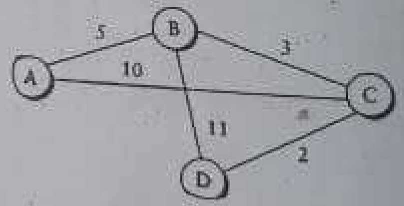
14. (a) What is learning bridges? How the incoming packets are forwarded to next hop by learning bridges? (6)

(b) Discuss the frame format of IEEE 802.11 wireless network. (5)

15. Consider 2 Ethernet LANs and a single host is connected using token ring. Both the Ethernets are connected to the token ring using router. In each Ethernet two hosts are attached. Draw the network diagram for the above connectivity. And if a host attached with Ethernet 1 wants to send data to host attached with Ethernet 2, identify the transmission route and the various frames transmitted through the route. State why this transmission requires network address along with physical address. (11)

Or

16. The above undirected weighted graph is the graphical representation of a network with 4 routers. Each node in the graph represents routers. Find the routing table of 'Node A" and "Node D" using link state routing (OSPF) protocol.



UNIT V

19. (a) Explain briefly about

(b) What is Ring topology

Or

20. Explain briefly about the correction techniques.

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B.Tech. DEGREE EXAMINATION, MAY 2018.

Fifth Semester

Computer Science and Engineering

COMPUTER NETWORKS

(2013 – 2014 onwards)

Time : Three hours　　　　Maximum : 75 marks

PART A — ($10 \times 2 = 20$ marks)

Answer ALL questions.

All questions carry equal marks.

1. State the advantage and limitations of Virtual Private Network.

2. Differentiate between single mode and multi mode fiber.

3. A bit string, 0111101111101111110, needs to be transmitted at the data link layer. Write the string actually transmitted after bit stuffing.

4. Give two reasons why networks might use an error-correcting code instead of error detection and retransmission.

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B.Tech. DEGREE EXAMINATION, APRIL/MAY 2016.

Fifth Semester

Computer Science and Engineering

COMPUTER NETWORKS

Time : Three hours — Maximum : 75 marks

PART A — ($10 \times 2 = 20$ marks)

Answer ALL questions.

All questions carry equal marks.

1. Name the two types of transmission technology.

2. What is the difference between a passive star and an active repeater in a fiber optic network?

3. Define hamming distance.

4. What is hub?

5. Define spanning tree.

6. Why traffic shaping approach is used in congestion?

7. What is Berkeley socket?

8. What are the two different types of multiplexing?

9. For what purpose the domain name system is used?

10. Write the difference between static web pages and dynamic web pages.

PART B — (5 × 11 = 55 marks)

Answer ALL questions, ONE from each Unit.

All questions carry equal marks.

UNIT I

11. Explain detail about the OSI reference model of computer network architecture.

Or

12. Explain briefly about the fiber optics.

UNIT II

13. Explain Error detection and correction briefly.

Or

14. Describe briefly about CSMA protocol with a diagram.

UNIT III

15. (a) Explain the services provided to transport layer.

(b) What is fragmentation? Explain briefly.

Or

16. Explain briefly about Internetworking and explain bow networks differ.

UNIT IV

17. How the transport layer protocol establishes and releases the connection?

Or

18. Explain briefly about TCP protocol and its management.

UNIT V

19. What is Email? Describe the architecture and services of Email.

Or

20. Explain the RSA algorithm.

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19. Discuss i and Corre

20. Write ela

(a) Addr

(b) Swit

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B.E. DEGREE EXAMINATION, APRIL/MAY 2017.

Fifth Semester

Computer Sciences and Engineering

COMPUTER NETWORKS

(2009-12 Batches)

Time : Three hours Maximum : 75 marks

PART A — ($10 \times 2 = 20$ marks)

Answer ALL questions.

All questions carry equal marks.

1. Write the differences between connection oriented service connectionless service.

2. What are the main two reasons for using layered protocols? Write one possible disadvantage of using layered protocols.

3. List the four types of framing methods used in data link layer.

4. Write the working principle of basic bit map protocol.

5. Compare virtual circuits and datagram.

6. Write the disadvantages of distance vector routing.

7. Draw a state diagram for simple connection management scheme.

8. What are the techniques used to restrict the packet lifetime?

9. Write the differences between MPEG and JPEG standard.

10. Write short notes on transposition techniques.

PART B — ($5 \times 11 = 55$ marks)

Answer ALL questions.

11. (a) Write short notes on coaxial cable. (4)

(b) Compare fiber optics and copper wires. (5)

(c) Define footprint and spot beams. (2)

Or

12. (a) Explain three forms of spread spectrum techniques with necessary diagram. (6)

(b) Discuss the functions of network and application layer of OSI reference model. (5)

13. (a) Mention name of the flow ... approached.

(b) Illustrate the error correcting codes wi... example.

Or

14. (a) List the key assumptions for dyn... channel allocation.

(b) Explain the working principle of ca... sense multiple access protocols.

15. (a) Define DAG.

(b) Discuss the working principle of dis... vector routing protocol and disadvantages.

Or

16. (a) Mention the reason for designing ... technique of tunneling and explain ... process of tunneling.

(b) Write the advantages and disadvantag... hierarchical addresses.

(c) Convert the IP address whose hexade... representation is C22F1582 to d... decimal notation.

7. What is a virtual circuit?

8. What are the features in OSPF?

9. Mention the types of error correcting methods.

10. What is the use of Switch?

PART B — (5 × 11 = 55 marks)

Answer ALL questions, ONE from each unit.

All questions carry equal marks.

UNIT I

11. What is meant by Access network? Explain the categories of Access network in detail? (11)

Or

12. Explain in detail about Internet protocol stack. (11)

UNIT II

13. Explain the Non-Persistent and Persistent Connections of HTTP. (11)

Or

14. Write in detail about RTCP. (11)

UNIT III

15. Explain the operation of the following in RDT : (11)

(a) GBN

(b) SR.

Or

16. (a) Write in detail about TCP segment structure. (6)

(b) Explain the case study of sequence number acknowledgement number. (5)

UNIT IV

17. Explain the architecture of Router. (11)

Or

18. (a) Discuss about ICMP. (6)

(b) Explain about the implementation of RIP. (5)

UNIT V

19. Write in detail about CSMA protocol. (11)

Or

20. Discuss about the Bridges. (11)

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B.Tech. DEGREE EXAMINATION, NOVEMBER 2015.

Fifth Semester

Computer Science and Engineering

COMPUTER NETWORK

(2009-2012 Batches)

Time : Three hours Maximum : 75 marks

PART A — (10 × 2 = 20 marks)

Answer ALL questions.

All questions carry equal marks.

1. What are the responsibilities of data link layer?

2. Discuss about the two interfaces provided by protocols.

3. What is the purpose of Domain Name System?

4. Name four factors needed for a secure network?

5. What is meant by quality of service?

6. What are data grams?

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B.Tech. DEGREE EXAMINATION, NOVEMBER 2018.

Fifth Semester

Computer Science and Engineering

COMPUTER NETWORKS

(2013-14 onwards batches)

Time : Three hours Maximum : 75 marks

PART A — ($10 \times 2 = 20$ marks)

Answer ALL questions.

All questions carry equal marks.

1. What is router?

2. Define signal to noise ratio. What is the unit for signal to noise ratio?

3. Write any two functions of data link layer.

4. What is hub?

5. Write short notes on multi destination routing.

6. Expand :

(a) OSPF

(b) BGP

7. Differentiate upward multiplexing and downward multiplexing.

8. What is meant by congestion Window?

9. Define cryptography.

10. Write the format of Resource Record in DNS.

PART B — (5 × 11 = 55 marks)

Answer ALL questions ONE from each Unit.

UNIT I

11. Discuss briefly about TCP/IP reference model of computer networks. (11)

Or

12. Explain the working of Fiber Optics transmission media. (11)

UNIT II

13. Describe briefly about Error detecting and correcting codes with examples. (11)

Or

14. (a) Explain the architecture of 802.11. (6)

(b) Write short notes on ALOHA protocol. (5)

UNIT III

15. (a) Explain the services provided to the Transport layer. (6)

(b) What is multicast routing? Explain. (5)

Or

16. (a) What is packet fragmentation? Explain. (6)

(b) Write short notes on Tunneling. (5)

UNIT IV

17. How the connection establishment can be done in transport layer? (11)

Or

18. (a) What is RPC? Explain briefly. (6)

(b) How will you regular the sending rate of information in transport layer? (5)

UNIT V

19. (a) Explain the architecture of E-mail. (6)

(b) Differentiate static web page and dynamic web page. (5)

Or